earlier_seen | value-weighted | Absolute loss surfaces
Expanding controls are deduplicated sidebands, not finite coordinates
Exploratory seen-period estimates; calibration/power deferred; no calibrated significance claim

Separate | A: matched PCA

PC count K	52	78	104	130	156	182	208	234	260	286	312	338	364	390	416	442	468	494	520
8	1.101	0.956	0.923	0.891	0.878	0.860	0.854	0.848	0.849	0.841	0.840	0.841	0.841	0.838	0.837	0.828	0.821	0.822	0.818
7	1.083	0.954	0.919	0.882	0.876	0.860	0.848	0.841	0.845	0.840	0.839	0.840	0.839	0.836	0.833	0.826	0.820	0.819	0.815
6	1.071	0.938	0.910	0.878	0.874	0.854	0.842	0.836	0.838	0.838	0.834	0.835	0.836	0.833	0.831	0.825	0.820	0.820	0.817
5	1.055	0.930	0.912	0.881	0.864	0.848	0.837	0.830	0.834	0.836	0.834	0.833	0.834	0.834	0.831	0.824	0.819	0.819	0.817
4	1.045	0.920	0.901	0.872	0.866	0.845	0.838	0.832	0.836	0.833	0.831	0.834	0.834	0.831	0.831	0.822	0.820	0.820	0.818
3	1.035	0.921	0.899	0.865	0.860	0.846	0.839	0.831	0.832	0.830	0.829	0.830	0.831	0.829	0.830	0.821	0.818	0.819	0.817
2	1.008	0.918	0.897	0.862	0.857	0.848	0.842	0.830	0.831	0.830	0.830	0.832	0.832	0.831	0.832	0.824	0.821	0.822	0.818
1	0.990	0.897	0.877	0.856	0.851	0.843	0.839	0.830	0.831	0.831	0.831	0.833	0.833	0.832	0.833	0.825	0.822	0.823	0.820

Coefficient-history pairs

Separate | B: expanding PCA

PC count K	52	78	104	130	156	182	208	234	260	286	312	338	364	390	416	442	468	494	520
8	1.152	1.002	0.947	0.900	0.879	0.860	0.853	0.850	0.850	0.845	0.842	0.843	0.842	0.839	0.839	0.829	0.825	0.826	0.823
7	1.147	0.990	0.938	0.892	0.874	0.855	0.849	0.844	0.846	0.841	0.838	0.840	0.839	0.837	0.836	0.826	0.822	0.823	0.820
6	1.108	0.967	0.928	0.886	0.869	0.852	0.846	0.840	0.842	0.838	0.836	0.837	0.836	0.833	0.833	0.823	0.819	0.820	0.817
5	1.080	0.958	0.925	0.877	0.860	0.845	0.839	0.834	0.838	0.835	0.834	0.837	0.836	0.835	0.834	0.825	0.821	0.823	0.820
4	1.071	0.950	0.914	0.873	0.858	0.844	0.837	0.832	0.835	0.831	0.831	0.834	0.832	0.831	0.831	0.823	0.820	0.821	0.819
3	1.055	0.930	0.901	0.864	0.851	0.840	0.834	0.828	0.831	0.829	0.828	0.830	0.830	0.828	0.828	0.820	0.817	0.817	0.815
2	1.041	0.931	0.902	0.868	0.856	0.846	0.840	0.831	0.832	0.832	0.831	0.833	0.834	0.833	0.834	0.826	0.822	0.823	0.820
1	0.988	0.895	0.876	0.854	0.849	0.842	0.838	0.829	0.830	0.830	0.830	0.832	0.832	0.831	0.833	0.825	0.822	0.822	0.820

Coefficient-history pairs

Shared control

PC count K	Exp
8	0.817
7	0.815
6	0.812
5	0.815
4	0.814
3	0.813
2	0.819
1	0.819

Coefficient-history pairs

Common | A: matched PCA

PC count K	52	78	104	130	156	182	208	234	260	286	312	338	364	390	416	442	468	494	520
8	0.993	0.893	0.871	0.858	0.858	0.846	0.838	0.829	0.830	0.827	0.827	0.830	0.830	0.829	0.828	0.820	0.816	0.816	0.814
7	0.988	0.891	0.873	0.853	0.860	0.847	0.836	0.827	0.830	0.827	0.828	0.830	0.831	0.829	0.826	0.820	0.816	0.816	0.813
6	0.986	0.887	0.868	0.852	0.859	0.842	0.834	0.824	0.827	0.827	0.826	0.828	0.828	0.828	0.825	0.820	0.817	0.817	0.814
5	0.977	0.889	0.869	0.854	0.850	0.839	0.831	0.822	0.825	0.826	0.826	0.828	0.828	0.830	0.828	0.822	0.818	0.818	0.815
4	0.982	0.884	0.866	0.850	0.853	0.838	0.833	0.825	0.828	0.826	0.827	0.830	0.829	0.829	0.828	0.821	0.819	0.819	0.817
3	0.977	0.890	0.874	0.849	0.852	0.841	0.834	0.825	0.826	0.824	0.824	0.827	0.827	0.827	0.827	0.820	0.818	0.818	0.816
2	0.967	0.895	0.876	0.851	0.852	0.845	0.839	0.825	0.825	0.826	0.826	0.829	0.828	0.829	0.829	0.822	0.820	0.820	0.817
1	0.963	0.882	0.866	0.847	0.845	0.838	0.835	0.826	0.827	0.827	0.827	0.830	0.830	0.831	0.831	0.824	0.821	0.822	0.819

Coefficient-history pairs

Common | B: expanding PCA

PC count K	52	78	104	130	156	182	208	234	260	286	312	338	364	390	416	442	468	494	520
8	1.014	0.926	0.889	0.859	0.853	0.843	0.837	0.830	0.830	0.828	0.828	0.831	0.830	0.830	0.829	0.822	0.819	0.820	0.817
7	1.012	0.922	0.887	0.857	0.852	0.841	0.835	0.827	0.829	0.827	0.826	0.829	0.828	0.828	0.828	0.820	0.818	0.818	0.815
6	0.983	0.903	0.876	0.851	0.847	0.836	0.830	0.821	0.824	0.823	0.823	0.826	0.825	0.824	0.824	0.816	0.814	0.814	0.811
5	0.992	0.908	0.881	0.849	0.844	0.834	0.827	0.819	0.823	0.823	0.823	0.828	0.826	0.826	0.827	0.819	0.816	0.817	0.814
4	0.998	0.907	0.878	0.849	0.846	0.836	0.830	0.822	0.825	0.824	0.825	0.829	0.826	0.826	0.826	0.819	0.817	0.817	0.815
3	0.989	0.896	0.872	0.844	0.842	0.834	0.829	0.821	0.823	0.823	0.824	0.827	0.826	0.827	0.826	0.818	0.816	0.816	0.814
2	0.998	0.912	0.886	0.856	0.852	0.843	0.836	0.825	0.826	0.827	0.826	0.829	0.829	0.830	0.830	0.823	0.820	0.821	0.817
1	0.962	0.880	0.864	0.845	0.844	0.837	0.834	0.825	0.826	0.826	0.826	0.829	0.830	0.831	0.831	0.823	0.821	0.821	0.818

Coefficient-history pairs

Shared control

PC count K	Exp
8	0.815
7	0.814
6	0.812
5	0.814
4	0.814
3	0.814
2	0.818
1	0.819

Coefficient-history pairs

Partial | A: matched PCA

PC count K	52	78	104	130	156	182	208	234	260	286	312	338	364	390	416	442	468	494	520
8	1.014	0.914	0.885	0.871	0.865	0.850	0.843	0.834	0.835	0.830	0.830	0.834	0.835	0.833	0.831	0.823	0.817	0.818	0.815
7	1.004	0.911	0.891	0.865	0.867	0.851	0.839	0.830	0.835	0.831	0.831	0.834	0.834	0.832	0.829	0.823	0.818	0.818	0.814
6	0.997	0.906	0.885	0.865	0.865	0.847	0.836	0.828	0.831	0.831	0.828	0.831	0.831	0.830	0.829	0.822	0.818	0.819	0.816
5	0.989	0.905	0.884	0.865	0.856	0.843	0.832	0.825	0.829	0.830	0.829	0.830	0.832	0.834	0.831	0.825	0.820	0.821	0.818
4	0.990	0.895	0.879	0.861	0.861	0.842	0.834	0.827	0.830	0.827	0.828	0.830	0.830	0.831	0.830	0.823	0.821	0.822	0.819
3	0.979	0.902	0.882	0.858	0.857	0.843	0.836	0.826	0.828	0.827	0.827	0.829	0.829	0.829	0.829	0.821	0.818	0.819	0.816
2	0.966	0.904	0.882	0.854	0.855	0.846	0.839	0.826	0.828	0.828	0.828	0.830	0.830	0.830	0.831	0.824	0.821	0.821	0.817
1	0.969	0.889	0.871	0.851	0.849	0.840	0.836	0.828	0.830	0.830	0.830	0.832	0.832	0.832	0.833	0.825	0.822	0.823	0.820

Coefficient-history pairs

Partial | B: expanding PCA

PC count K	52	78	104	130	156	182	208	234	260	286	312	338	364	390	416	442	468	494	520
8	1.029	0.947	0.907	0.874	0.865	0.850	0.845	0.836	0.837	0.833	0.833	0.837	0.836	0.835	0.835	0.825	0.822	0.824	0.820
7	1.035	0.941	0.904	0.871	0.863	0.849	0.841	0.831	0.833	0.831	0.829	0.834	0.832	0.832	0.831	0.822	0.820	0.821	0.817
6	1.001	0.917	0.895	0.864	0.855	0.843	0.835	0.827	0.828	0.827	0.827	0.828	0.829	0.828	0.826	0.818	0.814	0.817	0.814
5	1.006	0.919	0.897	0.861	0.852	0.840	0.832	0.824	0.827	0.827	0.826	0.830	0.830	0.830	0.830	0.822	0.818	0.821	0.817
4	1.011	0.919	0.889	0.856	0.852	0.840	0.833	0.826	0.829	0.828	0.828	0.832	0.830	0.829	0.830	0.822	0.819	0.821	0.817
3	1.002	0.904	0.880	0.848	0.846	0.838	0.830	0.822	0.827	0.827	0.826	0.829	0.829	0.829	0.828	0.820	0.817	0.818	0.814
2	1.010	0.929	0.895	0.857	0.855	0.845	0.837	0.828	0.831	0.832	0.831	0.833	0.833	0.833	0.833	0.826	0.822	0.823	0.820
1	0.966	0.887	0.869	0.849	0.847	0.839	0.835	0.827	0.829	0.829	0.829	0.831	0.831	0.832	0.832	0.825	0.822	0.823	0.820

Coefficient-history pairs

Shared control

PC count K	Exp
8	0.814
7	0.812
6	0.811
5	0.813
4	0.813
3	0.812
2	0.818
1	0.819

Coefficient-history pairs

Mean target MSE (squared log units)

K=0 own-history baseline: one reference per L, shared across modes/rules

Own control